EXERCISE 1 Solve the following by factorisation :

1. $9x^2 - 12x + 4 = 0$
2. $2x^2 - 13x - 7 = 0$
3. $3x^2 = 16x + 12$
4. $3x^2 + 5x = 2$
5. $2x^2 + 3 = 5x$
6. $3x^2 + 8x + 4 = 0$
7. $3x^2 = 10x + 8$
8. $4x^2 + 4x = 3$
9. $4x^2 = 11x + 3$
10. $12x^2 = 11x + 15$
11. $7x^2 + 6x = 1$
12. $15x^2 + 2x = 56$

EXERCISE 2 Solve the following by factorisation :

1. $4x^2 + 7x = 0$
2. $2x^2 - 11x = 0$
3. $x^2 - 5x + 6 = 0$
4. $6x^2 + 2x = 0$
5. $3x^2 = 8x$
6. $x^2 = 2x + 8$
7. $3x^2 - 7x = 0$
8. $9x = 6x^2$
9. $9 + x^2 = 6x$
10. $x^2 + x = 12$
11. $x^2 + 21 = 10x$
12. $x^2 + 8x = 33$

EXERCISE 3

Solve exactly for x :

1. $(x + 5)^2 = 2$
2. $(x - 8)^2 = 7$
3. $(x + 1)^2 + 1 = 11$
4. $(x + 6)^2 = -11$
5. $2(x + 3)^2 = 10$
6. $(2x + 1)^2 = 3$
7. $(x - 4)^2 = 8$
8. $3(x - 2)^2 = 18$
9. $(1 - 3x)^2 - 7 = 0$

EXERCISE 4 Solve exactly by completing the square :

1. $x^2 - 4x + 1 = 0$
2. $x^2 = 4x + 3$
3. $x^2 + 6x = 2$
4. $2x^2 + 4x + 1 = 0$
5. $x^2 + 6x + 2 = 0$
6. $x^2 + 6x + 7 = 0$

7. $x^2 = 2x + 6$
8. $x^2 + 10 = 8x$
9. $x^2 - 14x + 46 = 0$
10. $x^2 + 6x = -11$
11. $3x^2 = 6x + 4$
12. $2x^2 - 10x + 3 = 0$
13. $3x^2 + 12x + 5 = 0$
14. $5x^2 - 15x + 2 = 0$
15. $4x^2 + 4x = 5$

EXERCISE 5 Use the quadratic formula to solve exactly for x :

1. $x^2 - 4x - 3 = 0$
2. $x^2 + 6x + 7 = 0$
3. $x^2 + 1 = 4x$
4. $x^2 + 4x = 1$
5. $x^2 - 4x + 2 = 0$
6. $2x^2 - 2x - 3 = 0$
7. $(3x + 1)^2 = -2x$
8. $(x + 3)(2x + 1) = 9$

EXERCISE 6 By using the discriminant only, determine which of the following quadratic equations have rational roots which can be found by factorisation.

1. $6x^2 - 5x - 6 = 0$
2. $6x^2 - 47x - 8 = 0$
3. $2x^2 - 7x - 5 = 0$
4. $4x^2 - 3x + 2 = 0$
5. $3x^2 + 4x + 1 = 0$
6. $8x^2 + 2x - 3 = 0$

EXERCISE 7 By using the discriminant only, state the nature of the solutions of :

1. $x^2 + 7x - 3 = 0$
2. $5x^2 + 4x - 3 = 0$
3. $x^2 - 3x + 2 = 0$
4. $x^2 + x + 5 = 0$
5. $3x^2 + 2x - 1 = 0$
6. $16x^2 - 8x + 1 = 0$

EXERCISE 8 For each of the following quadratic equations, determine the discriminant Δ in simplest form and draw its sign diagram. Hence find the value(s) of m for which the equation has :
(i) a repeated root (ii) two distinct real roots (iii) no real roots.

1. $x^2 + 4x + m = 0$
2. $mx^2 + 3x + 2 = 0$
3. $mx^2 - 3x + 1 = 0$

EXERCISE 9

1. Using axes intercepts only, sketch the graphs of :

a. $y = (x-4)(x+2)$

b. $y = -(x-4)(x+2)$

c. $y = 2(x+3)(x+5)$

d. $y = -3x(x+4)$

e. $y = 2(x+3)^2$

f. $y = -\frac{1}{4}(x+2)^2$

2. State the equation of the axis of symmetry for each graph.

EXERCISE 10 Match each quadratic function with its corresponding graph.

1. $y = 2(x-1)(x-4)$

2. $y = -(x+1)(x-4)$

3. $y = (x-1)(x-4)$

4. $y = (x+1)(x-4)$

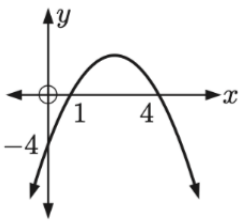
5. $y = 2(x+4)(x-1)$

6. $y = -3(x+4)(x-1)$

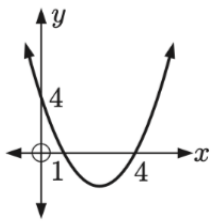
7. $y = -(x-1)(x-4)$

8. $y = -3(x-1)(x-4)$

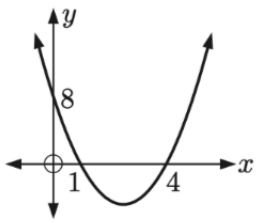
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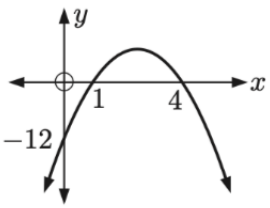
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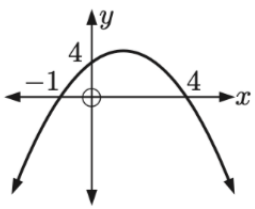
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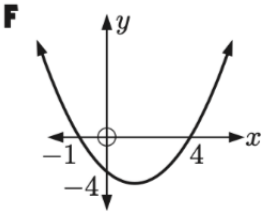
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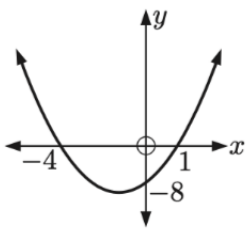
E



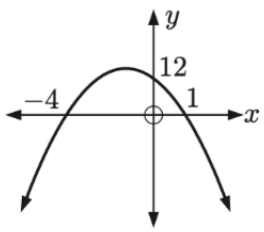
F



G



H



EXERCISE 11 Use the vertex, axis of symmetry, and y-intercept to graph :

1. $y = (x-1)^2 + 3$

2. $y = 2(x+2)^2 + 1$

3. $y = -2(x-1)^2 - 3$

4. $y = \frac{1}{2}(x-3)^2 + 2$

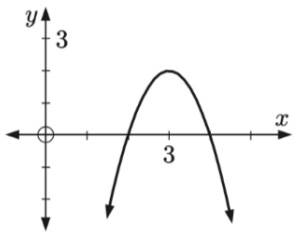
5. $y = -\frac{1}{3}(x-1)^2 + 4$

6. $y = -\frac{1}{10}(x+2)^2 - 3$

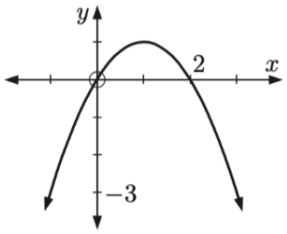
EXERCISE 12 Match each quadratic function with its corresponding graph :

1. $y = -(x+1)^2 + 3$
2. $y = -2(x-3)^2 + 2$
3. $y = x^2 + 2$
4. $y = -(x-1)^2 + 1$
5. $y = (x-2)^2 - 2$
6. $y = \frac{1}{3}(x+3)^2 - 3$
7. $y = -x^2$
8. $y = -\frac{1}{2}(x-1)^2 + 1$
9. $y = 2(x+2)^2 - 1$

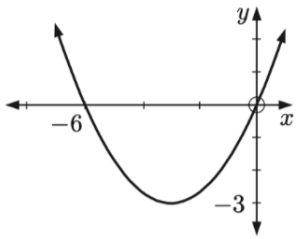
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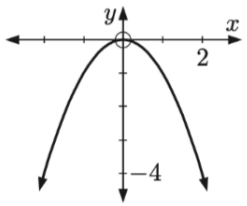
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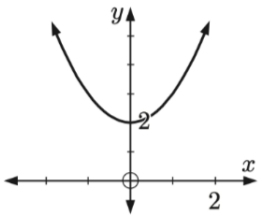
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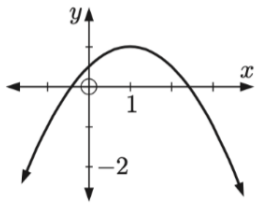
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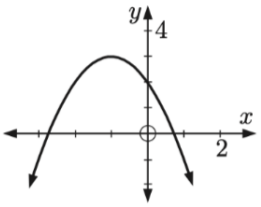
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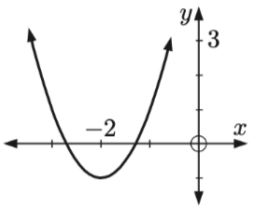
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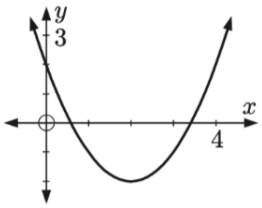
G



H



I



EXERCISE 13 Locate the turning point or vertex for each of the following quadratic functions :

1. $y = x^2 - 4x + 2$
2. $y = x^2 + 2x - 3$
3. $y = 2x^2 + 4$
4. $y = -3x^2 + 1$
5. $y = 2x^2 + 8x - 7$
6. $y = -x^2 - 4x - 9$
7. $y = 2x^2 + 6x - 1$
8. $y = 2x^2 - 10x + 3$
9. $y = -\frac{1}{2}x^2 + x - 5$

EXERCISE 14 Find the x-intercepts for :

1. $y = x^2 - 9$
2. $y = 2x^2 - 6$
3. $y = x^2 + 7x + 10$
4. $y = x^2 + x - 12$
5. $y = 4x - x^2$
6. $y = -x^2 - 6x - 8$
7. $y = -2x^2 - 4x - 2$
8. $y = 4x^2 - 24x + 36$
9. $y = x^2 - 4x + 1$
10. $y = x^2 + 4x - 3$
11. $y = x^2 - 6x - 2$
12. $y = x^2 + 8x + 11$

EXERCISE 15 For each of the following quadratics :

1. state the axis of symmetry
2. find the coordinates of the vertex
3. find the axes intercepts, if they exist
4. sketch the quadratic

1. $y = x^2 - 2x + 5$
2. $y = x^2 + 4x - 1$
3. $y = 2x^2 - 5x + 2$
4. $y = -x^2 + 3x - 2$
5. $y = -3x^2 + 4x - 1$
6. $y = -2x^2 + x + 1$
7. $y = 6x - x^2$
8. $y = -x^2 - 6x - 8$
9. $y = -\frac{1}{4}x^2 + 2x + 1$

EXERCISE 16 For each of the following quadratics :

1. Write the quadratic in the form $y = a(x - h)^2 + k$ without using technology.
2. State the coordinates of the vertex.
3. Find the y-intercept.
4. Sketch the graph of the quadratic.
5. Use technology to check your answers.

1. $y = 2x^2 + 4x + 5$
2. $y = 2x^2 - 8x + 3$
3. $y = 2x^2 - 6x + 1$
4. $y = 3x^2 - 6x + 5$
5. $y = -x^2 + 4x + 2$
6. $y = -2x^2 - 5x + 3$

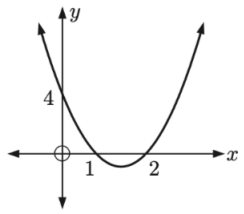
EXERCISE 17 Use the discriminant to determine the relationship between the graph and x-axis for :

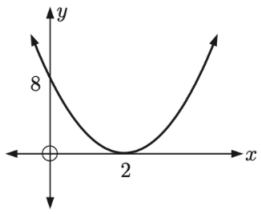
1. $y = x^2 + x - 2$
2. $y = x^2 - 4x + 1$
3. $y = -x^2 - 3$
4. $y = x^2 + 7x - 2$
5. $y = x^2 + 8x + 16$
6. $y = -2x^2 + 3x + 1$
7. $y = 6x^2 + 5x - 4$
8. $y = -x^2 + x + 6$
9. $y = 9x^2 + 6x + 1$

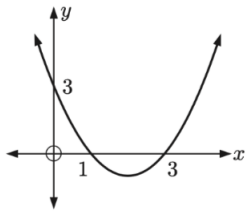
EXERCISE 18 Show that :

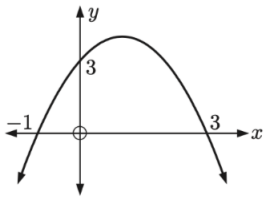
1. $x^2 - 3x + 6 > 0$ for all x
2. $4x - x^2 - 6 < 0$ for all x
3. $2x^2 - 4x + 7$ is positive definite
4. $-2x^2 + 3x - 4$ is negative definite

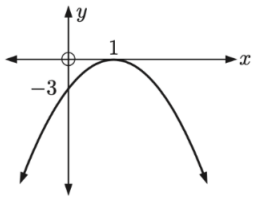
EXERCISE 19 Find the equation of the quadratic with graph :

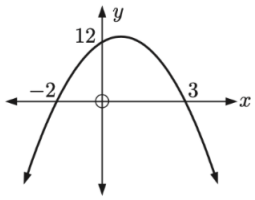
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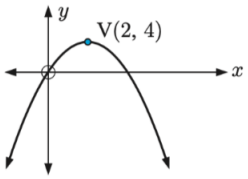
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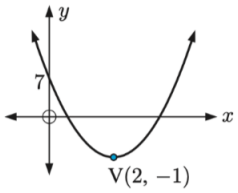
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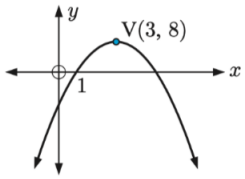
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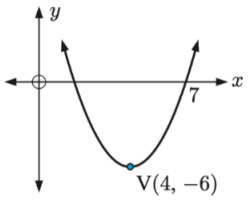
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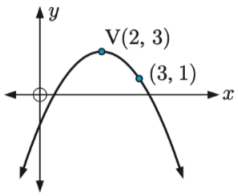
EXERCISE 20 If V is the vertex, find the equation of the quadratic function with graph :

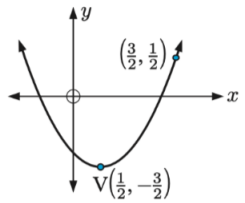
a 

b 

c 

d 

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f 

EXERCISE 21 Without using technology, find the coordinates of the point(s) of intersection of :

1. $y = x^2 - 2x + 8$ and $y = x + 6$
2. $y = -x^2 + 3x + 9$ and $y = 2x - 3$
3. $y = x^2 - 4x + 3$ and $y = 2x - 6$
4. $y = -x^2 + 4x - 7$ and $y = 5x - 4$

EXERCISE 22 The product of two consecutive odd numbers is 255. Find the numbers.

EXERCISE 23 The number of diagonals of an n-sided polygon is given by the formula

$$D = \frac{n}{2}(n-3).$$

A polygon has 90 diagonals. How many sides does it have ?

EXERCISE 24 The length of a rectangle is 4 cm longer than its width. The rectangle has area 26 cm^2. Find its width.

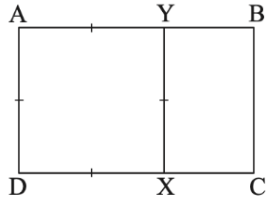
EXERCISE 25 The rectangle $ABCD$ is divided into a square and a smaller rectangle by $[XY]$ which is parallel to its shorter sides. The smaller rectangle $BCXY$ is *similar* to the original rectangle, so rectangle $ABCD$ is a **golden rectangle.**

The ratio $\frac{AB}{AD}$ is called the **golden ratio.**

Show that the golden ratio is

$$\frac{1+\sqrt{5}}{2}.$$

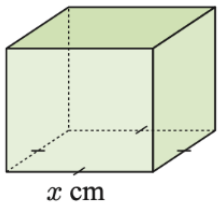
Hint : Let $AB = x$ units and $AD = 1$ unit.



EXERCISE 26 Two trains travel a 160 km track each day. The express travels 10 km h^{-1} faster and takes 30 minutes less time than the normal train. Find the speed of the express.

EXERCISE 27 A group of elderly citizens chartered a bus for \$160. However, at the last minute 8 of them fell ill and had to miss the trip. As a consequence, the other citizens had to pay an extra \$1 each. How many elderly citizens went on the trip ?

EXERCISE 28 A rectangular box has a square base with sides of length x cm. Its height is 1 cm longer than its base side length. The total surface area of the box is 240 cm^2.

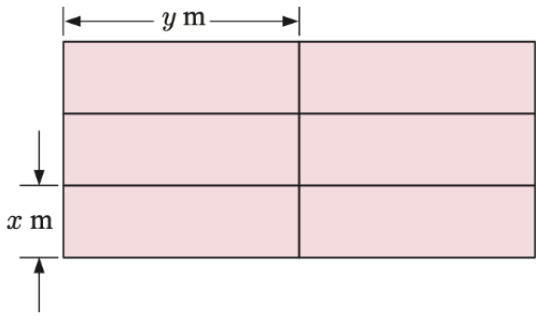


1. Show that the total surface area is given by

$$A = 6x^2 + 4x \text{ cm}^2.$$

2. Find the dimensions of the box.

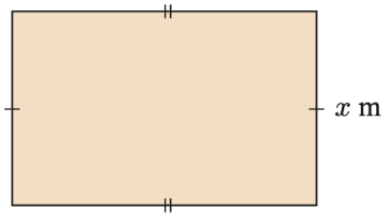
EXERCISE 29 A rectangular plot is enclosed by 200 m of fencing and has an area of A square metres. Show that :



1. $A = 100x - x^2$ where x m is the length of one of its sides.
2. The area is maximised if the rectangle is a square.

EXERCISE 30 Three sides of a rectangular paddock are to be fenced, the fourth side being an existing straight water drain. If 1000 m of fencing is available, what dimensions should be used for the paddock so that it encloses the maximum possible area ?

EXERCISE 31 1800 m of fencing is available to fence six identical pens as shown in the diagram.



1. Explain why $9x + 8y = 1800$.

2. Show that the area of each pen is given by

$$A = -\frac{9}{8}x^2 + 225x \text{ m}^2.$$

3. If the area enclosed is to be maximised, what are the dimensions of each pen ?